Shayan Amouzadeh

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RESEARCH INTERESTS

AI & Machine Learning: LLMs, NLP, multimodal speech-to-text, and information retrieval.
Software Engineering & Computer Systems: Automated testing and formal verification, high-performance and deterministic computing, computer graphics, and secure web architectures.

EDUCATION

B.Sc. in Computer Engineering (Software Engineering Specialization)

Semnan University, Faculty of Electrical and Computer Engineering | Semnan, Iran

Study period: Sep 2022 – Sep 2026 | **GPA (20-point scale):** 15.77/20 cumulative; 16.93/20 (last 2 years); 17.03/20 (final year); 18.03/20 across 21 selected core CS/CE courses.

Selected coursework: Programming Languages Design (19.50), Software Engineering (19.25), Operating Systems Lab (19.25), Linear Algebra (18.00), Data Structures & Algorithms (17.50).

UNDERGRADUATE THESIS & CAPSTONE PROJECT

Developer & System Architect — Intelligent Real Estate CRM Platform

B.Sc. Thesis: Design & Implementation of an Intelligent Real Estate Management System | Grade: 19.50/20 | Advisor: Dr. Morteza Dorrigiv, Semnan University

Live project: amlak-e-man.ir | GitHub: Real-Estate-File-Manager-Project

- **Architecture:** Built and deployed a multi-tenant CRM with PHP/PDO, a 7-table MySQL schema, 17 REST handlers, and a vanilla-JavaScript PWA; added runtime migrations, role-based access, composite indexes, and 15-second change-token synchronization.
- **Applied AI & security:** Added human-reviewed Persian voice entry using Whisper-large-v3 and OpenRouter LLM extraction, plus property matching and Neshan maps; implemented tenant-bound HMAC-SHA256 authentication, rate limiting, and 47 automated tests.

SELECTED TECHNICAL PROJECTS

IEEE 754 Single-Precision Floating-Point Unit

C++17 · CMake · Hardware Arithmetic Verification | GitHub

- Built a constexpr binary32 model for addition, subtraction, multiplication, and division. Differential testing passed 32,026,992 operand-pair checks with zero mismatches across GCC, Clang, AppleClang, and MSVC CI.

Deterministic Graphics Pipeline & CPU Software Rasterizer

C++17 · C99 · OpenGL 1.1 · FreeGLUT | GitHub

- Refactored an animated OpenGL scene and built a deterministic C99 rasterizer with 49 GL/GLUT entry points. Added pixel-exact regression tests across three operating systems, resolved an ARM64 FMA discrepancy, and removed 605 legacy GL errors per frame.

SELECTED TECHNICAL PROJECTS (CONTINUED)

Bare-Metal ARMv7-A VGA Framebuffer Renderer

ARMv7-A Assembly · Python · Unicorn CPU Emulation | GitHub

- Implemented an RGB565 renderer for the DE1-SoC memory-mapped VGA framebuffer. Unicorn CI verified output with zero out-of-bounds writes; optimization reduced instructions by 67% (353,286 to 116,201).

End-to-End Box-Office Scraping & Content Recommender Pipeline

Python · Pandas · BeautifulSoup4 · Streamlit | GitHub

- Built a Persian box-office pipeline covering 2013–2024 (1392–1403 SH): 686 films, 52 columns, and 43 genre features; added dual Top-K recommendations and 14 automated tests.

Exact-Optimal Fractional Knapsack & DP Benchmark Suite

C++17 · Python Benchmarking | GitHub

- Developed a header-only O(n log n) solver with a dynamic-programming reference; validated 328,582 checks with zero failures and benchmarked 1.86 ms versus 26.38 s at n=20,000.

PROFESSIONAL EXPERIENCE

Computer Engineer & IT Specialist | Mar 2025 – Present (Full-Time)

Khakab Sazeh Ghomes Co. | Semnan, Iran

- Maintain company computers, internal networks, hardware/software, and user accounts; handle installation, troubleshooting, and day-to-day IT support. Build Access/Excel reports and automate financial/engineering billing, scheduling, and resource tracking with Microsoft Project.

Game Developer & Software Engineering Intern | Jul 2025 – Dec 2025

Sargarmi Sazan Aseman Omid (Game Development Studio) | Iran

- Built Unity/C# gameplay systems with Entitas ECS and uGUI/TextMeshPro; profiled and optimized builds. Also developed an open-source Unity 6 directory UI with LINQ-powered search and sorting. GitHub project

TECHNICAL SKILLS

Programming: C++17, C, Python 3, ARMv7-A Assembly, C#, PHP 8, JavaScript, SQL, HTML5/CSS3

AI & data: LLM extraction, Whisper STT, Persian NLP, Pandas, BeautifulSoup4, Streamlit, WER/F1 evaluation

Systems & verification: IEEE 754, differential FPU testing, deterministic rasterization, Unicorn emulation, memory-mapped I/O

Web, graphics & tools: REST APIs, PWA, Neshan Leaflet, MySQL/MariaDB/SQLite, PDO, HMAC-SHA256/JWT, OpenGL/FreeGLUT, Unity 6/Entitas ECS, Git/Plastic SCM, GitHub Actions, CMake, Docker, Pytest, Ruff, Puppeteer

LANGUAGES

Persian: Native | **English:** Advanced working proficiency